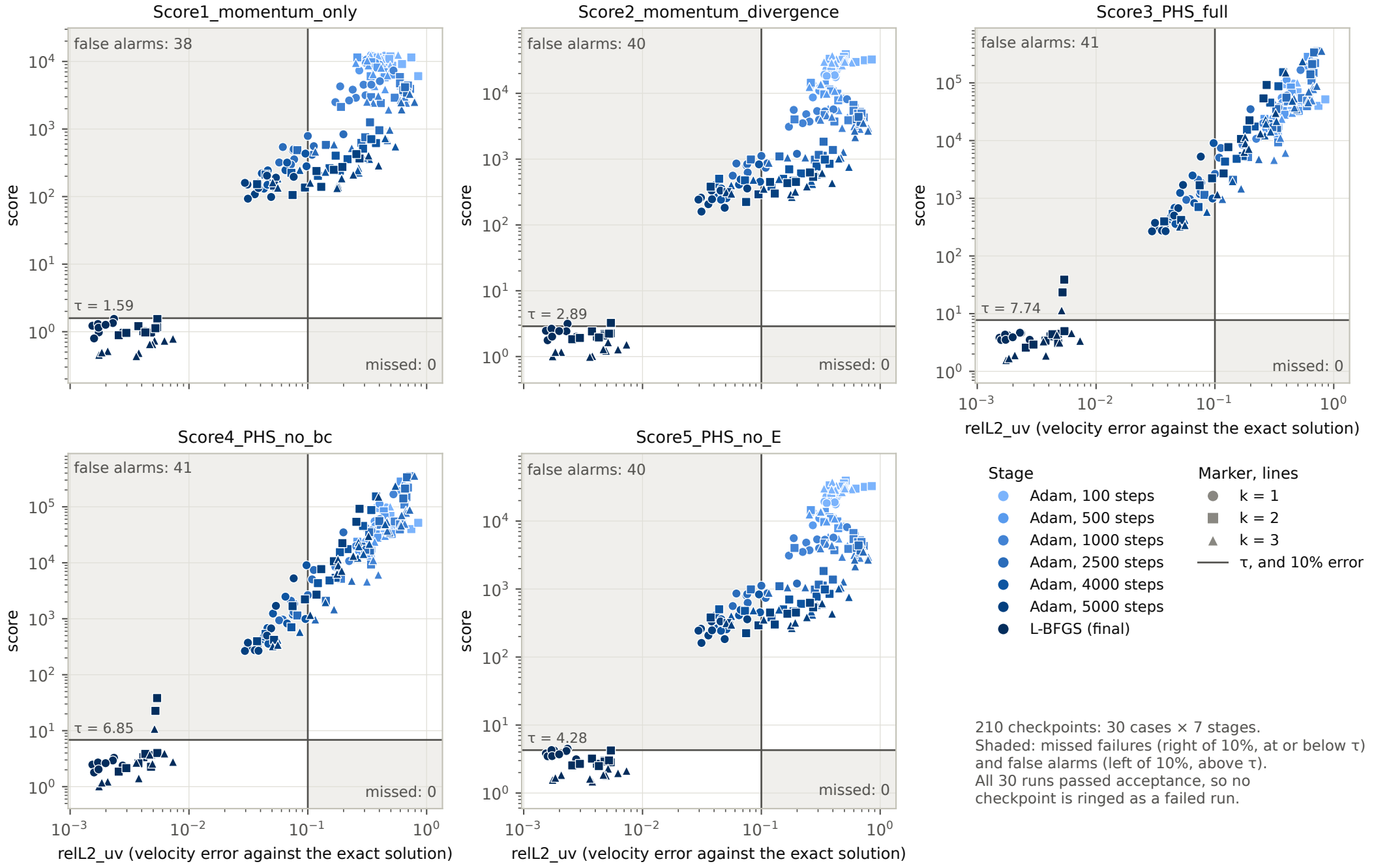

Checkpoint scores against the true velocity error (frozen calibration, flagged above τ)
Score1_momentum_only
false alarms: 38
τ = 1.59
missed: 0
Score2_momentum_divergence
false alarms: 40
τ = 2.89
missed: 0
Score3_PHS_full
false alarms: 41
τ = 7.74
missed: 0
Score4_PHS_no_bc
false alarms: 41
τ = 6.85
missed: 0
Score5_PHS_no_E
false alarms: 40
τ = 4.28
missed: 0
score
relL2_uv (velocity error against the exact solution)
Stage
Adam, 100 steps
Adam, 500 steps
Adam, 1000 steps
Adam, 2500 steps
Adam, 4000 steps
Adam, 5000 steps
L-BFGS (final)
Marker, lines
k = 1
k = 2
k = 3
τ, and 10% error
210 checkpoints: 30 cases × 7 stages.
Shaded: missed failures (right of 10%, at or below τ)
and false alarms (left of 10%, above τ).
All 30 runs passed acceptance, so no
checkpoint is ringed as a failed run.